



When the machine is working as synchronous generator, for low field current I_{f1} , *under excitation* ($E_f = E_{f1}$), the stator current ($I_a = I_{a1}$) is large and leading. The stator current is minimum ($I_a = I_{a1}$) and at unity power factor for the field current I_{f2} ($E_f = E_{f2}$) which is called normal excitation. For larger field current I_{f3} , *over excitation* ($E_f = E_{f3}$) the stator current ($I_a = I_{a3}$) is large and lagging.

When the machine is working as generator, it supplies a lagging power factor current when over-excited and leading power factor current when under excited. When the machine is working as motor, it draws a leading power factor current when over-excited and draws a lagging power factor current when under excited.

If the synchronous machine is not transferring any power but is simply floating on the infinite bus, the power factor is zero, that is, the stator current either leads or lags the stator voltage by 90° . The magnitude of the stator current changes as the field current is changed, but the stator current is always reactive. Looking for the machine terminals, the machine behaves as a variable inductor or capacitor as the field current is changed. An unloaded synchronous machine is called a synchronous condenser and may be used to regulate the receiving end voltage of a long power transmission line.

PROCEDURE

Make the connections as per the diagram.

Precautions :

- i) Keep dc motor field rheostat in minimum position
- ii) Keep alternator field potential divider in the minimum voltage position
- iii) Keep DPDT, DPST, TPST1, TPST2; switches open
- iv) Keep the load on DC side in off position.

Switch on the supply to the dc motor (DPDT switch in position 1-1'). Start the motor using the 3 point starter and increase the speed to the synchronous speed by varying the motor field rheostat.

Now, switch on the supply to the alternator field and vary the potential divider so that the generated voltage is nearly equal to rated value. Close the TPST switch and note down the 3 phase supply voltage. Adjust the potential divider and make the generator terminal voltage equal to the 3 phase supply voltage. Now, the lamps will flicker in sequence. If the lamps are flickering uniformly i.e. All the lamps become dim or bright simultaneously (phase sequence is wrong), then interchange the two terminals of the 3 phase supply voltage after switching off the TPST and DPST switches. If the flickering is so fast, the motor field rheostat is adjusted very slightly so that the frequency of flickering is convenient and the synchronization switch is closed when two lamps show maximum brightness and the third dark. Synchronization is over. Now, the wattmeter shows zero reading.